CANAN KARAKOÇ

School of Biological Sciences, Georgia Institute of Technology, 310 Ferst Dr., Atlanta, GA 30332, US
ckarakoc3@gatech.edu | www.ckarakoc.com | github.com/canankarakoc

*Digital version includes external links.

EDUCATION

Ph.D. in Biology, *magna cum laude* — *2013 – 2019*
University of Leipzig , HIGRADE Graduate School — *Leipzig, Germany*

Primary advisers: Hauke Harms and Antonis Chatzinotas

Mentors: Adam T. Clark and Viktoriia Radchuk

Thesis title: "Context Dependency of Community Dynamics: Predator-Prey Interactions Under Ecological Disturbances"

M.Sc. in Global Change Ecology — *2009 – 2012*
University of Bayreuth , Elite Network Bavaria (ENB) — *Bayreuth, Germany*
Advisers: Björn Reineking, Steffen Kolb, George Wang, Detlef Weigel

M.Sc. in Biology — *2005 – 2008*
Cumhuriyet University — *Sivas, Turkey*

B.Sc. in Biology (Minor: Molecular Biology) — *2000 – 2004*
Akdeniz University — *Antalya, Turkey*

PROFESSIONAL EXPERIENCE

Brown Lab Biological Sciences, Georgia Institute of Technology
The Center for Microbial Dynamics and Infection (CMDI) — 2024 – Ongoing
Postdoctoral researcher — *Atlanta, GA, US*

- Microbial dynamics and infection, host-pathogen interactions, disease dynamics, antimicrobial resistance (AMR).
- Adviser: Sam Brown

Lennon Lab Department of Biology, Indiana University — 2021 – 2024
Postdoctoral researcher — *Bloomington, IN, US*

- GEMS (Genomics and Eco-evolution of Multi-scale Symbioses) trainee
- Microbial ecology and evolution, bioenergetics, dormancy.
- Adviser: Jay T. Lennon

Helmholtz Centre for Environmental Research–UFZ
German Centre for Integrative Biodiversity Research (iDiv) — 2019 – 2021
Postdoctoral researcher — *Leipzig, Germany*

- yDiv trainee
- Evolutionary ecology, species interactions, fluctuations, AMR.
- Advisers: Antonis Chatzinotas and Stan Harpole

UFZ 2013 – 2018
Research assistant *Leipzig, Germany*
· Experimental community ecology. WG Microbial Interactions Ecology

iDiv 2015 – 2016
Guest research assistant *Leipzig, Germany*
· Experimental community ecology. WG Experimental Interactions Ecology

University of Thessaly 2016 & 2014
Guest research assistant, DAAD Scholarship *Larissa, Greece*
· Applied microbial ecology. Department of Biochemistry and Biotechnology

Max Planck Institute for Developmental Biology 2011
Intern, ENB travel grant *Tübingen, Germany*
· Evolutionary ecology. Department of Molecular Biology

Technical University of Munich 2010
Intern, ENB travel grant *Munich, Germany*
· Microbial ecology. Department of Soil Ecology.

University of Bayreuth 2010 – 2012
Technical assistant *Bayreuth, Germany*
· Field/lab work. University of Bayreuth, Department of Soil Physics; Department of Biogeography (EVENT); Department of Plant Physiology (TERRECO) and Agroecosystem Research.

Cumhuriyet University 2005 – 2008
Research & teaching assistant *Sivas, Turkey*
· Applied microbial ecology. Department of Molecular Biology and Genetics.

Antalya State Hospital 2003
Intern *Antalya, Turkey*
· Laboratories of Microbiology, Immunology & Biochemistry.

AWARDS

2025 Spring Postdoctoral Research Fellowship Award program of the Cystic Fibrosis Foundation (CFF). *Eco-evolutionary processes govern clinical responses to antibiotics and the failure of susceptibility testing in people with CF. ∼ $150,000.*

iDiv FlexPool internal funding for postdoctoral research.

German Academic Exchange Service (DAAD). Two months travel scholarship for research in Greece.

HIGRADE (PhD), **yDiv** (PhD) and **GEMS** (postdoc) traineeships with research and travel support.

Swedish Royal Academy — Travel award for winter school in Sweden.

Elite Network Bavaria — Internships, courses, conferences, research and tuition fee.

PUBLICATIONS

Published

First author:

- **Karakoç, C.**, Shoemaker, W. R., and Lennon, J. T. (2026). Evolutionary bioenergetics of sporulation. *Proceedings of the National Academy of Sciences*, 123(6):e2524274123. doi.org/10.1073/pnas.2524274123 Code

- O'Sullivan, T.*, **Karakoç, C.***, Wollein-Waldetoft, K., and Brown, S. P. (2025). Risk of death during acute infection is accelerating across diverse host-pathogen systems and consistent with multiple models of host-pathogen interaction. *Equal contribution. *mSphere*, pages e00953–24. doi.org/10.1128/msphere.00953-24 Code

- **Karakoç, C.**, Clark, A. T., and Chatzinotas, A. (2020). Diversity and coexistence are influenced by time-dependent species interactions in a predator–prey system. *Ecology Letters*, 23(6). doi.org/10.1111/ele.13500

- **Karakoç, C.** (2019). Context dependency of community dynamics: Predator-prey interactions under ecological disturbances. *Ph.D. Thesis*, Leipzig University. https://nbn-resolving.org/urn:nbn:de:bsz:15-qucosa2-341500

- Sendek, A.*, **Karakoç, C.***, Wagg, C., Domínguez-Begines, J., Martucci de Couto, G., Van der Heijden, M. G., Naz, A. A., Lochner, A., Chatzinotas, A., Klotz, S., Gómez-Aparicio, L., and Eisenhauer, N. (2019). Drought modulates interactions between arbuscular mycorrhizal fungal diversity and barley genotype diversity. *Equal contribution. *Scientific Reports*, 9(1):1–15. doi.org/10.1038/s41598-019-45702-1

- **Karakoç, C.**, Radchuk, V., Harms, H., and Chatzinotas, A. (2018). Interactions between predation and disturbances shape prey communities. *Scientific Reports*, 8:2968. doi.org/10.1038/s41598-018-21219-x

- **Karakoç, C.**, Singer, A., Johst, K., Harms, H., and Chatzinotas, A. (2017). Transient recovery dynamics of a predator–prey system under press and pulse disturbances. *BMC Ecology*, 17:13. doi.org/10.1186/s12898-017-0123-2

- **Karakoç, C.** (2012). Population response to fluctuating temperature regimes – an analysis with a model microorganism. *M.Sc. Thesis*, University of Bayreuth

Other:

- Clark, A. T., Shoemaker, L. G., Arnoldi, J.-F., Barabás, G., Germain, R., Godoy, O., Hallett, L., **Karakoç, C.**, Saavedra, S., and Schreiber, S. J. (2026). A practical guide to characterising ecological coexistence. *Biological Reviews*, 101(1):195–220. doi.org/10.1111/brv.70079

- Glidden, C. K., **Karakoç, C.**, Duan, C., Jiang, Y., Beechler, B., Jabbar, A., and Jolles, A. E. (2023). Distinct life history strategies underpin clear patterns of succession in microparasite communities infecting a wild mammalian host. *Molecular Ecology*, 32(13):3733–3746. doi.org/10.1111/mec.16949

- Zhao, Q., Van den Brink, P., Xu, C., Wang, S., Clark, A., **Karakoç, C.**, Sugihara, G., Widdicombe, C. E., Atkinson, A., Matsuzaki, S., Shinohara, R., He, S., Wang, Y. X. G., and De Laender, F.

(2023). Relationships of temperature and biodiversity with stability of natural aquatic food webs. *Nature Communications*, 14(3507). doi.org/10.1038/s41467-023-38977-6

- Jurburg, S. D., Buscot, F., Chatzinotas, A., Chaudhari, N. M., Clark, A. T., Garbowski, M., Grenié, M., Hom, E. F. Y., **Karakoç, C.**, Marr, S., Neumann, S., Tarkka, M., van Dam, N. M., Weinhold, A., and Heintz-Buschart, A. (2022). The community ecology perspective of omics data. *Microbiome*, 10(225). doi.org/10.1186/s40168-022-01423-8

- Clark, A., Mühlbauer, L. K., Hillebrand, H., and **Karakoç, C.** (2022). Measuring stability in ecological systems without static equilibria. *Ecosphere*, 13(12):e4328. doi.org/10.1002/ecs2.4328

- Clark, A., Arnoldi, J.-F., Zelnik, Y., Barabás, G., Hodapp, D., **Karakoç, C.**, König, S., Radchuk, V., Donohue, I., Huth, A., Jacquet, C., de Mazancourt, C., Mentges, A., Nothaaß, D., Shoemaker, L., Taubert, F., Wiegand, T., Wang, S., Chase, J., Loreau, M., and Harpole, S. (2021). General statistical scaling laws for stability in ecological systems. *Ecology Letters*, 24(7):1474–1486. doi.org/10.1111/ele.13760

- Saraiva, J. P., Worrich, A., **Karakoç, C.**, Kallies, R., Chatzinotas, A., Centler, F., and Nunes da Rocha, U. (2021). Mining synergistic microbial interactions: A roadmap on how to integrate multi-omics data. *Microorganisms*, 9(4). doi.org/10.3390/microorganisms9040840

- Ozbayram, E. G., Akyol, C., Ince, B., **Karakoç, C.**, and Ince, O. (2018). Rumen bacteria at work: bioaugmentation strategies to enhance biogas production from cow manure. *Journal of Applied Microbiology*, 124(2):491–502. doi.org/10.1111/jam.13668

Publications under review and accepted

- **Karakoç C.**, O'Sullivan T., Gurney J., Martignoni M.M., Wollein Waldetoft K. and Brown S.P. (2026). Host health links infection history to survival during acute *Pseudomonas aeruginosa* infection. Under review *Proceedings B.* Code

- Martignoni M.M., Bordenstein S.R., **Karakoç C.**, Tyson R.C., Brown S.P., Garnier J. (2026). An eco-evolutionary consumer-resource theory of host-microbe symbioses. In press *Journal of Evolutionary Biology*.

First author works close to submission

- **Karakoç C.**, Goldstein R., and Brown S.P. **Host–parasite plasticity drives coevolutionary mismatches and transient immunopathology**. We examine host–pathogen coevolutionary dynamics using a game theoretical approach where players develop evolving strategies. Code

- **Karakoç C.**, Stecenko A., Diggle S., and Brown S.P. **Rethinking antibiotic susceptibility as a population-level, eco-evolutionary measurement**. We analyzed longitudinal antimicrobial resistance (AMR) data of a cohort of Cystic Fibrosis patients to discuss the effect of different sources of noise on the AMR inferences and the failure of treatments.

- **Karakoç C.**, Behringer M., Schoolmaster Jr. D.R. and Lennon J.T. **Mutation accumulation under extreme energy limitation.** We explored adaptive mutations occurring in *Bacillus subtilis* that engage dormancy after three years of starvation. Code

- **Karakoç C.**, Clark A.T., Hines J., Rall B.C., Latz E., Harpole S.W., Hiltunen T. and Chatzinotas A. **Antibiotic resistance under random and correlated environments in a predator-prey metacommunity**. We constructed a bacteria-phage metacommunity system to explore antibiotic and phage resistance trade-off under temporally random and autocorrelated antibiotic regimes. Code

PRESENTATIONS

Talks

- Invited Talk (2026). Antibiotic-driven resistance in *Pseudomonas aeruginosa* follows nonlinear, state-dependent dynamics in people with cystic fibrosis. North American Cystic Fibrosis Conference, Atlanta, GA.
- Invited Talk (2026). Rethinking resistance: Signal, noise & heterogeneity in antimicrobial susceptibility testing of cystic fibrosis *Pseudomonas aeruginosa*. Cystic Fibrosis Foundation Research Conference, Colorado Springs, CO.
- Contributed talk (2025). Disentangling host, pathogen, and interaction effects on infection-induced mortality. Evolution, Athens, GA.
- Contributed talk (2023). Evolution of complex traits through the lens of bioenergetics. ESA Annual Meeting, Portland, OR.
- Contributed talk (2020). Diversity and coexistence are influenced by time-dependent species interactions in a predator-prey system. ESA Annual Meeting, virtual.
- Contributed talk (2019). Diversity and stability are directly linked to fluctuating species interactions in a predator-prey system. GfÖ, Münster, Germany.
- Contributed talk (2015). Understanding community assembly mechanisms through integrative approaches, EEF-SiTE - Ecology at the Interface, 2015, Rome, Italy.
- Contributed talk (2014). Understanding the role of species interactions under environmental change: Microbial model systems as tools in ecological theory. YoMo Workshop - Ecological modeling across disciplines, Hann. Münden, Germany
- Invited talk (2014/2016). Patterns and processes under environmental fluctuations: Experiments with microbial model systems. University of Thessaly, Department of Biochemistry and Biotechnology, Larissa, Greece.

Posters

- Poster (2026). Antibiotic-driven resistance in *Pseudomonas aeruginosa* follows nonlinear, state-dependent dynamics in people with cystic fibrosis. North American Cystic Fibrosis Conference, Atlanta, GA.
- Poster (2026). A multi-method statistical framework for quantifying antibiotic-driven resistance evolution in *Pseudomonas aeruginosa* among people with cystic fibrosis. The annual Cystic Fibrosis Foundation Research Conference, Colorado Springs, CO.

- Poster (2026). From patterns to processes: Meta-analysis and experiments reveal cumulative damage drives infection-induced mortality. EEID Ecology and Evolution of Infectious Disease Conference, Blacksburg, VA.
- Poster (2025). Dynamics of death: Why and when a host dies during infection. Gordon Research Conference, Microbial Population Biology, Andover, NH.
- Poster (2023). Evolution of survival through the lens of bioenergetics. AbGradCon, San Diego, CA.
- Poster (2021). Community constraints in adaptation to stressors. ESA Annual Meeting, virtual.
- Posters (2018). (a) Resolving Complex Microbial Community Dynamics: A causality analysis with microbial model systems. (b) Impact of Nutrient Levels and Stoichiometry on Microbial Freshwater Community and Functioning in Microcosm Experiments. ISME17, Leipzig, Germany.
- Poster (2014). Transient dynamics of trophically interacting species after disturbance. HETEROCLIM: The response of organisms to climate change in heterogeneous environments, Loches, France.

PROFESSIONAL SKILLS

Computer programs

Proficient	Version control, R, tidyverse, Python, Bash, LaTeX
Familiar with	Mathematica, NetLogo, Java, Julia, QGIS/ArcGIS, ImageJ

Wet lab/field

Proficient	Microcosms consisting of viruses, bacteria, protozoa
Familiar with	Nematodes, wax moth, grassland & green house experiments

Illustration/Science communication

	Pen & paper, Inkscape, Adobe Illustrator/InDesign Procreate, Affinity Designer

Languages

English	Fluent speaking & writing
German	Fluent speaking & writing
Turkish	Native

ACADEMIC MENTORING & TEACHING

Primary supervision

- Undergraduate research thesis (2026 – ongoing). Antimicrobial resistance in people with cystic fibrosis. Oliva E. Georgia Institute of Technology.
- Undergraduate lab training/research credit (2025 – ongoing). Microbiology, biochemistry, experimental evolution. Elisi B., Chloe L., and Pratyusha S. Georgia Institute of Technology.

- M.Sc. Bioinformatics (2025 – 2026). Evolutionary game theoretical models to understand host-pathogen co-evolution, Karan G., Georgia Institute of Technology.
- Undergraduate summer research (2025). Wet-lab methods in microbial ecology and evolution, Abhiram K., Georgia Institute of Technology.
- Undergraduate research thesis (2024 – 2025). Accelerating mortality driven by pathogen burden and host damage. Charlotte W., Georgia Institute of Technology.
- Undergraduate research experience (2024 – 2025). Testing biofilm formation and antimicrobial activity using synthetic polymers. Jackson L., Georgia Institute of Technology.
- Research project (Summer 2023), Microbial division of labor under ecological disturbances. Michalle M., Indiana University Summer STEM Research Program.
- B.Sc Research experience (2023 – 2024), Microcosms, experimental evolution. Emily C., Indiana University.
- Summer internship (2022). Stability of metabolic exchange and dormancy, Melih Ç., Middle East Technical University.
- B.Sc. project (2020 – 2021). Effect of environmental noise on antibiotic and bacteriophage resistance evolution, Klara-Isabell G., Leipzig University.
- B.Sc. project (2020 – 2021). Fitness costs of antibiotic resistance in various environments, Joanna S., Leipzig University.
- Internship (2018 – 2019) and Master Thesis (2019–2020). Evolutionary rescue in complex communities, Alla K., Leipzig University.
- Internship (2014). Predator–prey interactions under disturbances, Jana H., University of Kassel.

Mentoring

- High school project (2021 – 2022). Complexity Effects on Structural Stability: Using Protist Microcosms and Mathematical Modeling to Navigate Realism in Theoretical Ecology, Sylvia G., OPRFHS IRDI, Chicago, IL.
- B.Sc. project (2021). Effect of environmental noise on microbial evolution, Philipp K., Leipzig University.
- PhD chapter (2019 – 2020). Mechanisms promoting co-existence of blood-borne parasites in African buffalo, Caroline G., Oregon State University.
- PhD project (2018 – 2020). Microbial communities of amphibian skin microbiomes, spread of pathogenic chytrid fungus, Adriana C., University of Toulouse.
- PhD project (2018 – 2021). Microbial communities and their interactions across trophic levels in mountain lakes, Judit L., Leipzig University.

Teaching

- Undergraduate course (Spring 2025, 2026) BIOL 3400: Introduction to Mathematical Biology, Georgia Institute of Technology.
- Graduate course (2021 – 2023). Quantitative Biodiversity, Indiana University.
- Practical research training (2022 – 2023). Microbiology graduate program rotation students, Biology undergraduates, Indiana University.
- Literature seminar (2020 – 2021). Microbial Ecology, Leipzig University.
- Practical training (2018 – 2021). Measuring microbial diversity, experimental evolution, R for data science, Leipzig University.
- Practical courses (2005 – 2008). General biology, Genetics, Molecular Genetics, Biochemistry, Animal Physiology, Microbiology, Introduction to Molecular Biology, Molecular Cell Biology, Cumhuriyet University.

· Course materials (2018 – 2025). Activities, worksheets and R scripts from four of these courses: github.com/canankarakoc/teaching-materials

RELEVANT ACTIVITIES

Symposium: Incorporating Dormancy and Rarity to Predict Community Dynamics and Stability Under Environmental Change *2023*
Organizer *ESA Annual Meeting 2023, Portland, OR*

Course: Origin of Life *2022*
Participant *Complexity Explorer, Santa Fe Institute*

Workshop: Bioinformatics *2022*
Participant *GEMS, Urbana–Champaign, IL*

Workshop: Trait-Based Eco-Evolutionary Modeling *2019*
Participant *Leipzig, Germany*

Workshop: Filling in gaps in global understanding of ecological stability and coexistence *2019*
Invited participation *Leipzig, Germany*

Workshop: an introduction to Bayesian statistics *2019*
Participant, FlexPool travel grant *Münster, Germany*

Course: Introduction to regression models with spatial and temporal correlation R-INLA *2018*
Participant, Highland Statistics Ltd., UFZ DEVELOP training grant *Leipzig, Germany*

Workshop: Eco-evolutionary dynamics in experimental microbial communities *2018*
UFZ Controlling Chemicals' Fate invited speaker (Prof. Teppo Hiltunen) grant *Leipzig, Germany*

Minisymposium: Experimental evolution & community dynamics *2018*
Participant, FlexPool travel grant *Tvärminne, Finland*

Winter school: Marine evolution-patterns and processes, Centre for Marine Evol. Bio. *2011*
Participant, Swedish Royal Academy of Sciences travel award *Tjärnö, Sweden*

Modelling the fate of microbes in aquatic habitats and assessment of their associated risks *2010*
Participant, ENB travel grant *Vienna, Austria*

Other graduate school activities *2013-2018*
Courses, soft skill trainings Link to file *Leipzig, Germany*

Other graduate school activities *2009-2012*
Courses, soft skill trainings Link to file *Bayreuth, Germany*

SERVICE

Peer reviews for Grant Proposals
· National Science Foundation, Division of Environmental Biology (2023)

Scientific Committee, Award/Proposal Review
· Georgia Tech Undergraduate Research Spring Symposium Judge (2026–ongoing)

- Georgia Tech Undergraduate Research Opportunities Salary Award Proposal Review (2025–ongoing)
- Ecological Society of America Early Career Paper Award Review (2024–ongoing)

Administrative

- Leadership (2026–ongoing). Georgia Institute of Technology Postdoc association. Faculty Liaison, and Belonging and Outreach Teams.
- Leadership (2022–2024). Indiana University Postdoc association Career Development Board.

Editorial

- Community Editor in Frontiers in Ecology and Evolution. Speciality section Population, Community, and Ecosystem Dynamics.

Peer reviews for Journals (appx. 5-10/year)

- Nature Communications, Nature Computational Science, Nature Sustainability, Scientific Reports, Ecology Letters, Ecology, Biology Letters, Communications Biology, eLife, mSystems, mSphere, Frontiers in Microbiology, PLoS Biology, mBio, Methods in Ecology and Evolution, The American Naturalist, Applied and Environmental Microbiology, Frontiers in Ecology and Evolution, Oikos, PCI Ecology, PeerJ, Int. Journal of Appl. Comp. Math.

Memberships

- Response Diversity Network (RDN). Current.
- American Association for the Advancement of Science (AAAS). Current.
- American Society for Microbiology (ASM). Current.
- American Society of Naturalists (ASN). Current.
- Ecological Society of America (ESA). Past.

Media interviews

- Article (2026). Hibernating bacteria can invade hospitals and survive outer space – but their high energy bill is a weakness that scientists can strike. The Conversation US. https://doi.org/10.64628/AAI.t3jkrh5gr
- Interview (2018). Ökosystemforschung im Labor. Norddeutscher Rundfunk NDR.

Outreach

- Public outreach (2025). Atlanta Science Festival.
- Science Olympiad Mentor (2022–2023). Bloomington South High School, Bloomington, IN.
- Event organization (2022). Future faculty preparation conference, Indiana University.
- Session organization (2022). "Life of a postdoc", Graduate conference, Indiana University.
- Community outreach (2021, 2022). Bacterial viruses, Science Fest, Indiana University.
- Social development (2020). Better working culture, UFZ.
- Volunteer (2013). Federation of Microbiological Societies-FEMS Meeting, Leipzig, Germany.
- Nature conservation activities (2000-2005) Biodiversity monitoring, Doga (BirdLife International partner), Turkey.
- Voluntary teaching (2004-2005). English, arts. Educational volunteers foundation of Turkey.
- Science philosophy and ethics (2004). Workshop, panel and public survey. Akdeniz University.
- Astronomy seminar series for non-astronomers (2000-2003) Physics department, Akdeniz University.